📘 Introduction

Customer churn is a critical challenge for businesses in the telecommunications industry, as retaining existing customers is significantly more cost-effective than acquiring new ones. Churn refers to the phenomenon where customers discontinue their relationship with a service provider. Understanding the patterns and factors contributing to customer churn is essential for developing effective retention strategies.

In this project, we analyse the **Telco Customer Churn dataset**, which contains demographic, account, and service-related information about customers of a telecom company. Each entry indicates whether a customer has churned or not, making it suitable for a **binary classification problem**. The dataset includes features such as:

- Customer tenure and contract type
- Monthly and total charges
- Internet service and add-ons
- Demographics like gender and senior citizen status
- Payment methods and paperless billing

By leveraging machine learning models, we aim to:

- Identify key factors influencing churn,
- Build predictive models to classify churn risk,
- Evaluate and compare model performance,
- Suggest actionable insights for customer retention.

To address the challenge of customer attrition, predictive analytics plays a crucial role in enabling telecom companies to take proactive measures. This project utilizes a real-world dataset from a telecom company, containing over 7,000 customer records with demographic, service usage, and account information. Given the class imbalance inherent in churn data—where fewer customers leave than stay—techniques like SMOTEENN are employed to enhance model fairness. By applying various supervised machine learning models, including Random Forest, Logistic Regression, XGBoost, and Gradient Boosting, the study aims to identify key drivers of churn and build a robust predictive model. The ultimate goal is to help telecom businesses **anticipate churn** and **proactively engage at-risk customers**, thereby improving customer satisfaction and reducing revenue loss. These insights can support customer retention strategies, allowing the business to offer targeted incentives and improve service delivery to high-risk customers.

Customer Data Structure

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No
7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No

rows × 21 columns

The table above represents a snapshot of the Telco Customer Churn dataset, which contains customer-level data collected by a telecommunications company. Each row corresponds to a unique customer, identified by the customerID column. The dataset includes a mix of demographic, service usage, and account information.

Key attributes visible in the sample include:

- **Demographics**: gender, SeniorCitizen (binary indicator), Partner, and Dependents, which help describe the customer's household and age profile.
- **Account Tenure**: tenure indicates the number of months the customer has been with the company.
- **Service Features**: These include whether the customer has a PhoneService, MultipleLines, the type of InternetService (DSL, Fiber optic, or None), and additional services such as OnlineSecurity, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies.

These features are essential for identifying patterns related to customer churn. For instance, customers with short tenure or without value-added services may be more likely to churn. The full dataset contains 21 columns, which are used for exploratory data analysis and building predictive machine learning models.

🔍 Data Understanding

The dataset consists of **7,043** customer records and **21** features, including both categorical and numerical variables. Each row represents an individual customer, and the dataset captures key attributes that may influence churn behavior.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

Data Summary:

- Total Entries: 7,043
- No Missing Values: All columns have 7,043 non-null entries, indicating that there is no missing data.
- Memory Usage: Approximately 1.1 MB

Column Breakdown:

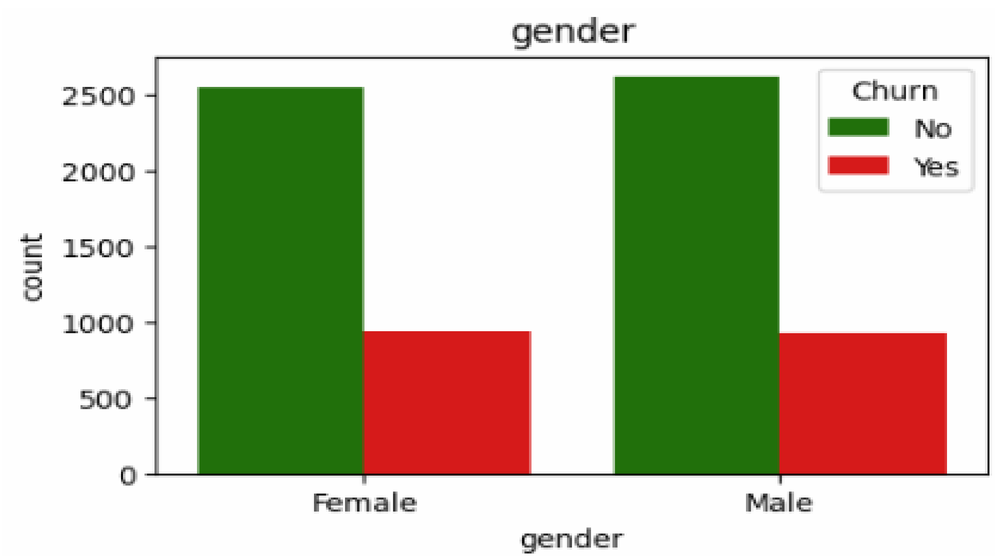
- Identifiers:
  - customerID: Unique identifier for each customer.
- Demographic Features:
  - gender, SeniorCitizen, Partner, Dependents
- Service-related Features:

- PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies

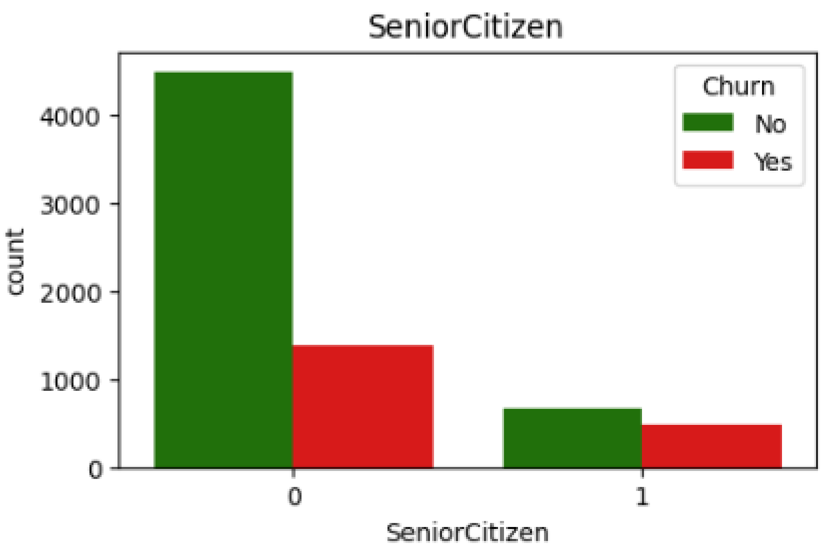
- **Account Information**:
    - Contract, PaperlessBilling, PaymentMethod
- **Usage & Charges**:
    - tenure (number of months with the company),
    - MonthlyCharges, TotalCharges
- **Target Variable**:
    - Churn: Indicates whether the customer has left the company (Yes) or not (No).

This step is crucial as it helps understand the structure, types, and completeness of the dataset before proceeding to data preprocessing and modeling.

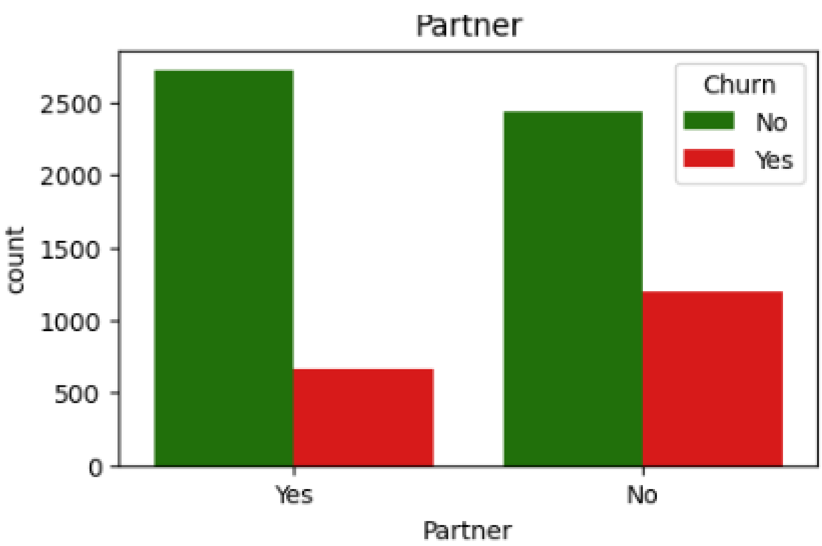
📊 Data Visualization



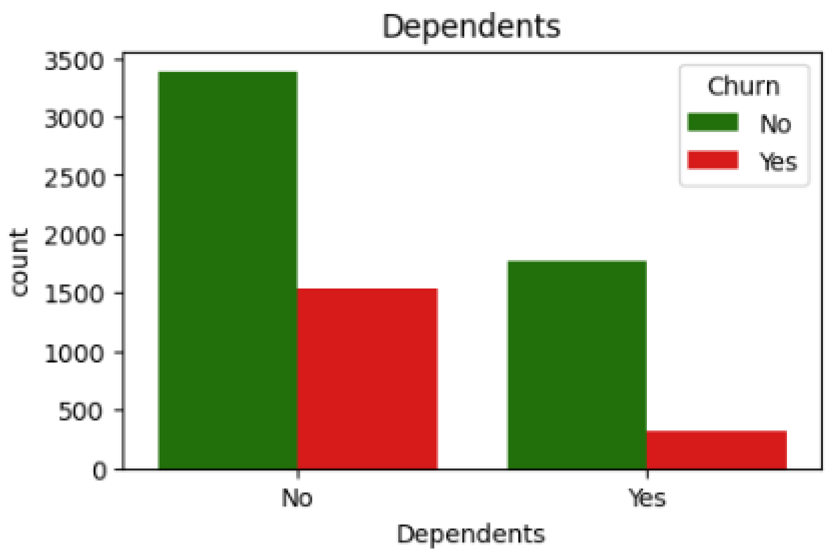
The bar chart shows the distribution of customer churn based on gender. Both male and female customers have a similar churn rate, with over 2,500 customers from each gender not churning and around 900 churning. This indicates that gender does not significantly influence customer churn in this dataset.



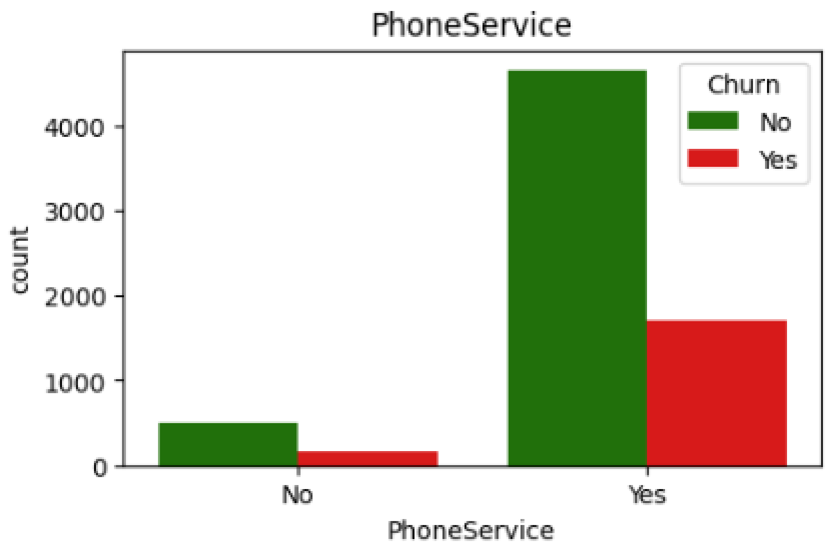
The bar chart shows the distribution of customer churn based on senior citizen status. Customers who are not senior citizens (0) have a significantly higher overall count and also more churn cases than senior citizens (1). However, the churn rate among senior citizens appears slightly higher proportionally, suggesting age may have some influence on churn behavior.



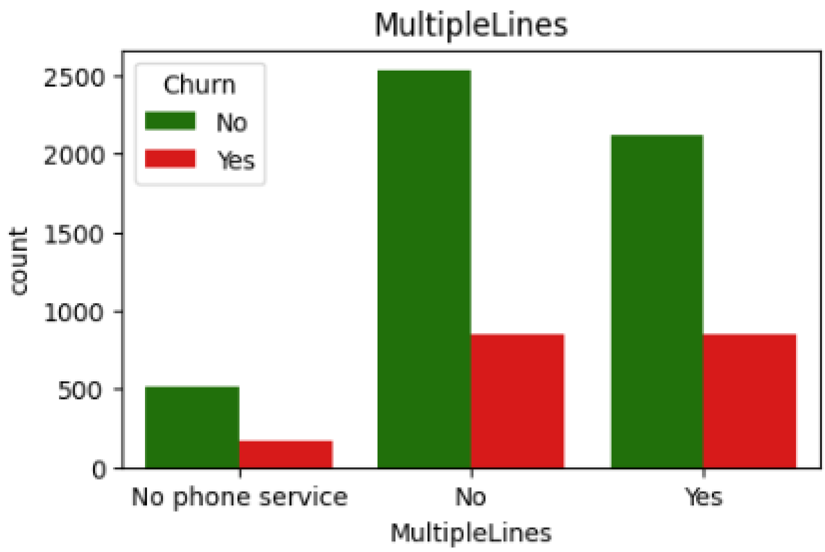
The chart illustrates customer churn based on whether the customer has a partner. Customers without a partner show a noticeably higher churn count compared to those with a partner. This suggests that having a partner may be associated with a lower likelihood of customer churn.



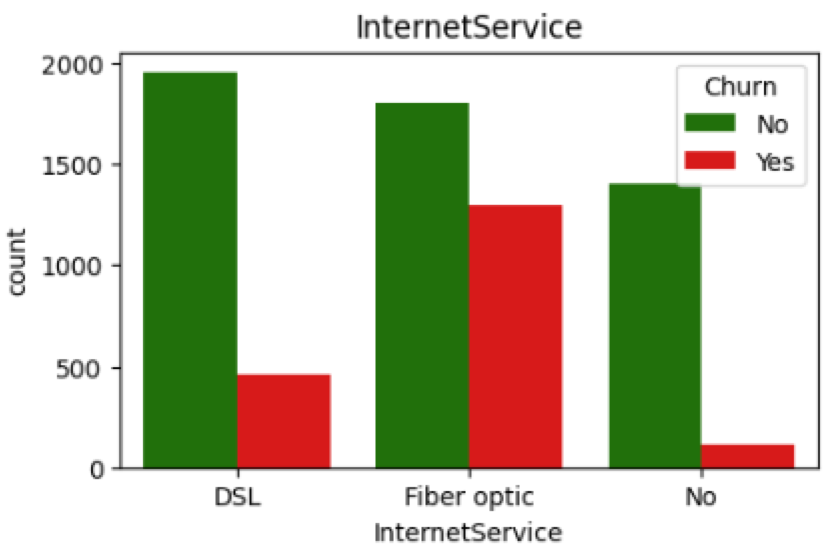
The bar chart illustrates customer churn based on whether customers have dependents. Customers without dependents show a significantly higher churn rate compared to those with dependents. This indicates that having dependents may contribute to greater customer retention.



The graph shows the relationship between phone service subscription and customer churn. Customers with phone service have both higher overall numbers and higher churn counts compared to those without phone service. However, the churn rate appears proportionally similar, indicating that having phone service alone may not be a strong predictor of churn.

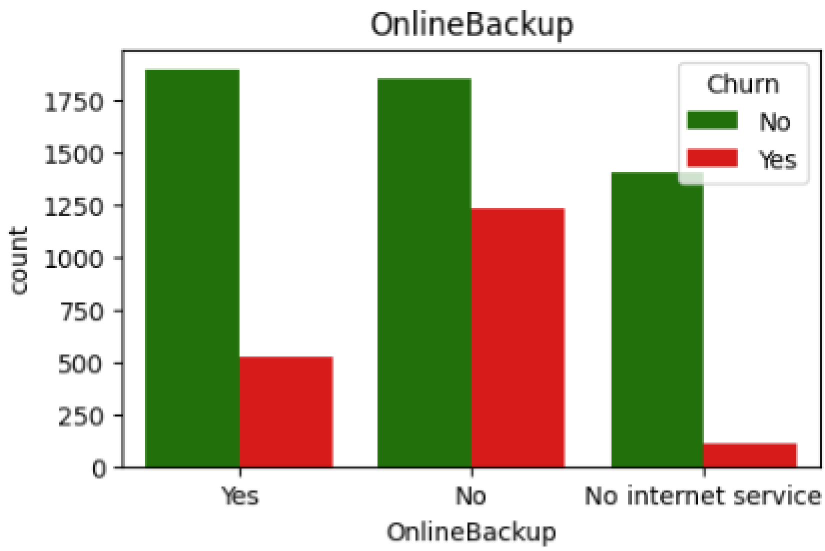


The graph shows churn distribution across different categories of the **MultipleLines** feature. Customers without multiple lines or with multiple lines both exhibit churn, but those without multiple lines are slightly more in number. The churn rate is comparatively lower among those with "No phone service", indicating they are less likely to churn.

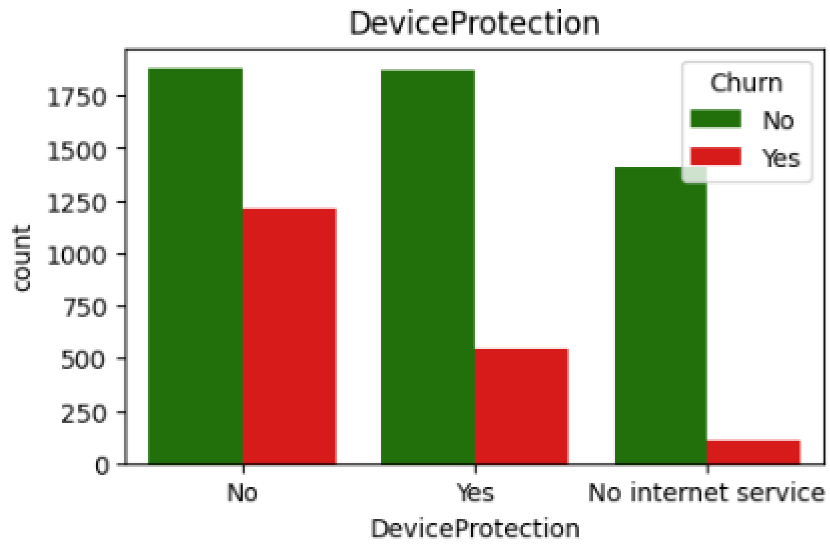


The graph illustrates the relationship between **Internet Service type** and customer churn. Customers using **Fiber optic** have the highest churn rate compared to DSL and those without internet service. Notably, customers with **no internet service** show the lowest churn, indicating a more stable segment.

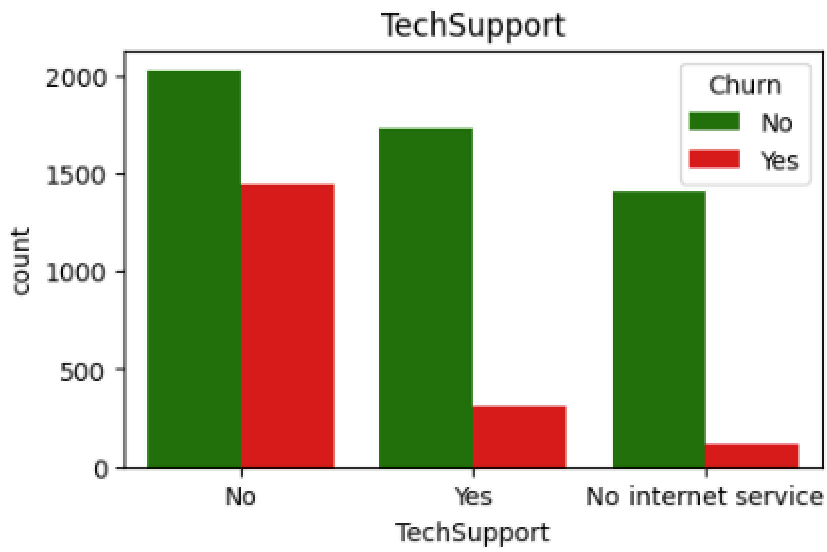
The graph shows that customers **without Online Security** have a significantly higher churn rate compared to those who have it. Customers with **Online Security services** are less likely to leave, suggesting a possible link between security features and customer retention. Those with **no internet service** have the lowest churn, similar to previous trends.



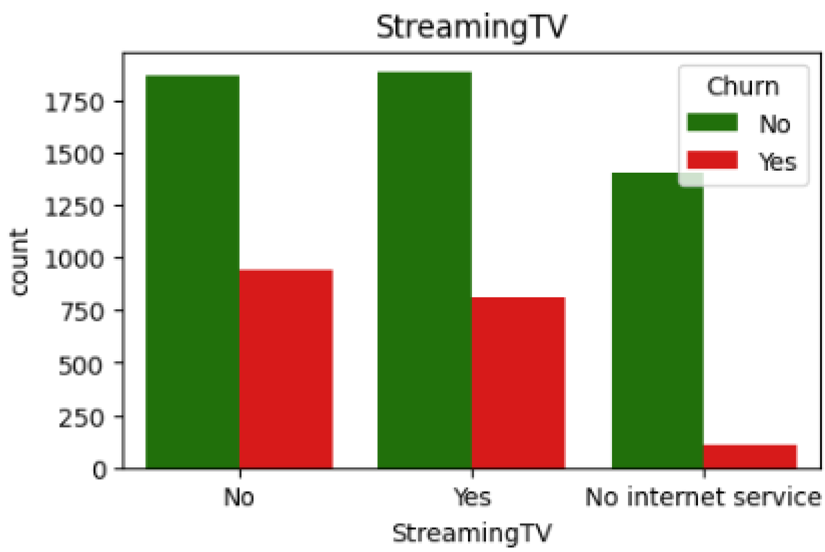
The chart shows that customers **without Online Backup** are more likely to churn than those who have the service. Customers who use **Online Backup** have noticeably lower churn counts, indicating better retention. Those with **no internet service** again show the least churn activity, aligning with earlier patterns.



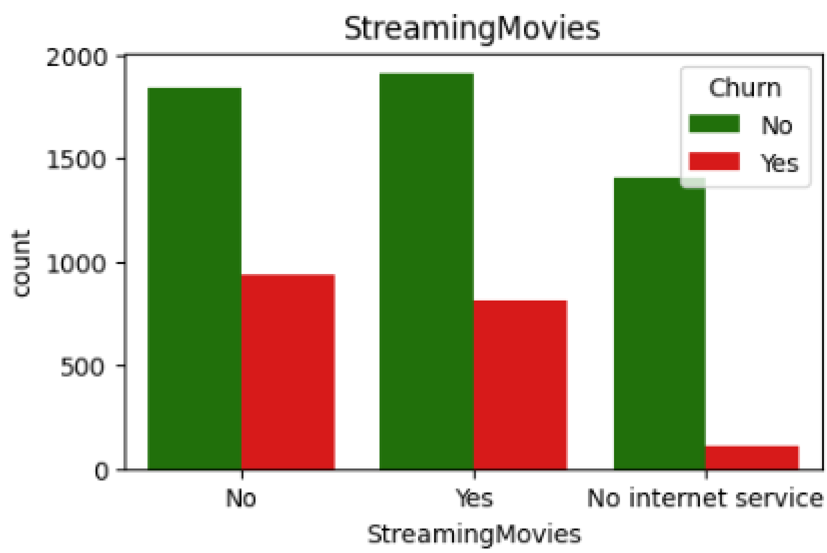
The chart reveals that customers **without Device Protection** have a significantly higher churn rate than those who have it. Customers with **Device Protection** show improved retention, suggesting this service may enhance customer satisfaction. Again, churn is lowest among those with **no internet service**, possibly due to limited engagement.



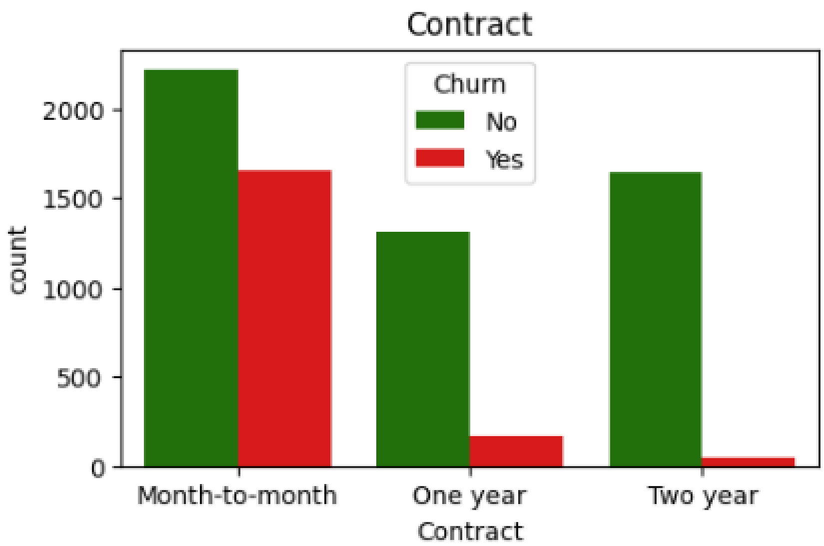
The graph shows that **customers without Tech Support** have a much higher churn rate compared to those who have it. Providing Tech Support appears to play a key role in retaining customers. As seen before, churn is lowest among users with **no internet service**, likely due to limited service interactions.



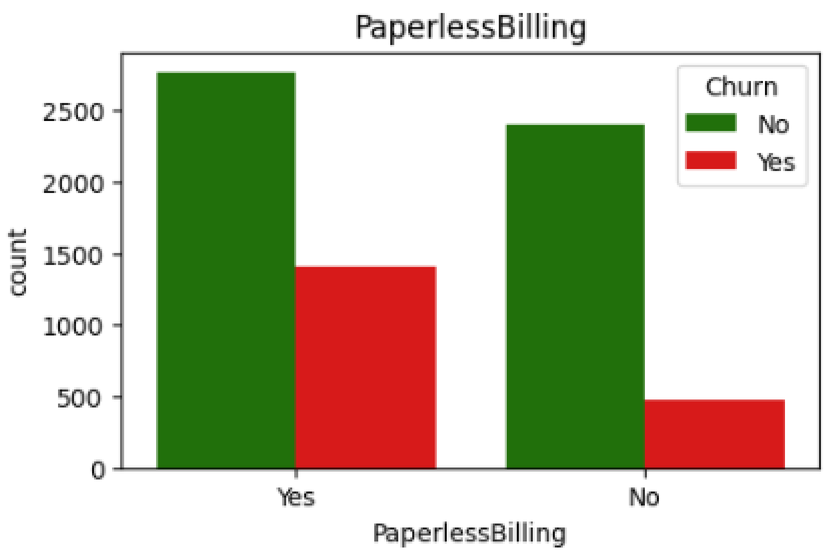
The graph shows that **churn rates are relatively similar** for customers regardless of whether they use Streaming TV or not. However, those with **no internet service** have the lowest churn, likely due to fewer services being active. This suggests that Streaming TV availability alone doesn't strongly influence churn decisions.



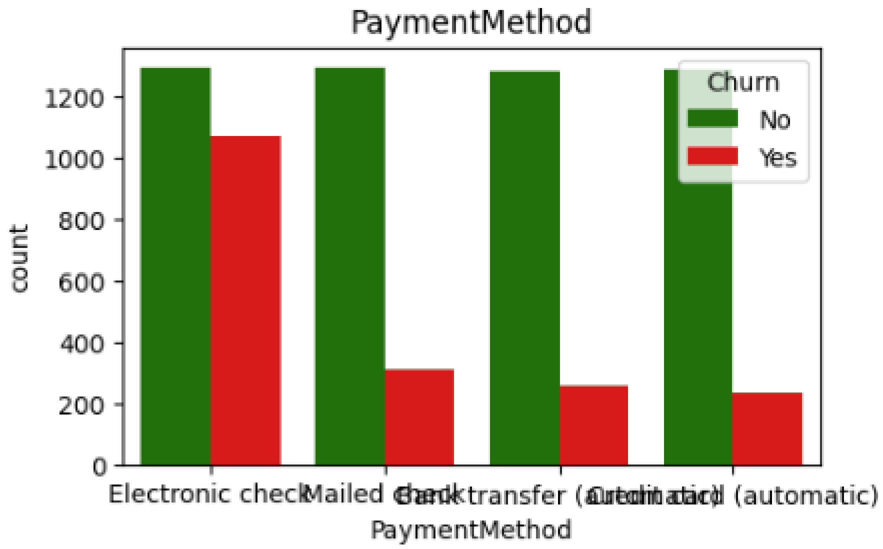
The graph indicates that churn rates are fairly balanced between customers who do and do not use **Streaming Movies**. However, customers with **no internet service** show the lowest churn, likely due to fewer services being subscribed. This implies that **Streaming Movies** usage does not significantly impact churn behavior on its own.



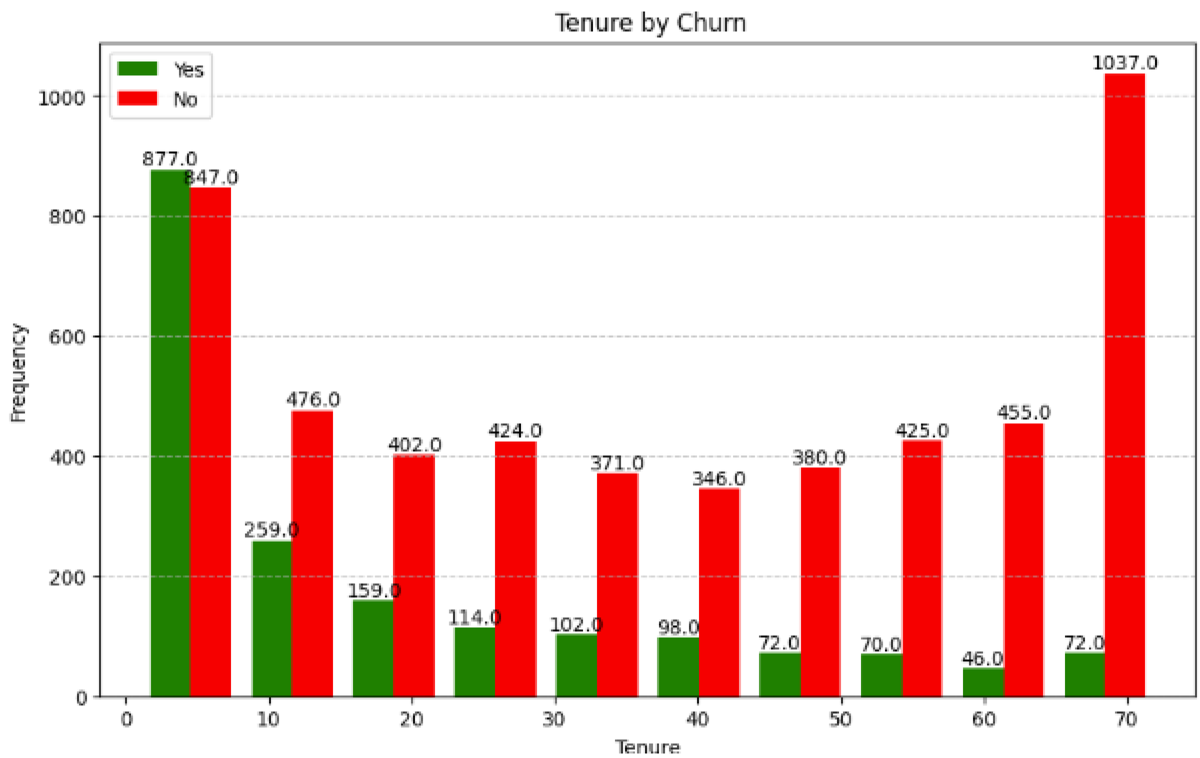
The graph shows that customers on **month-to-month contracts** have a significantly higher churn rate compared to those on **one-year** or **two-year contracts**. Churn is lowest among **two-year contract holders**, indicating greater customer retention. This suggests that **long-term contracts are effective in reducing churn**.



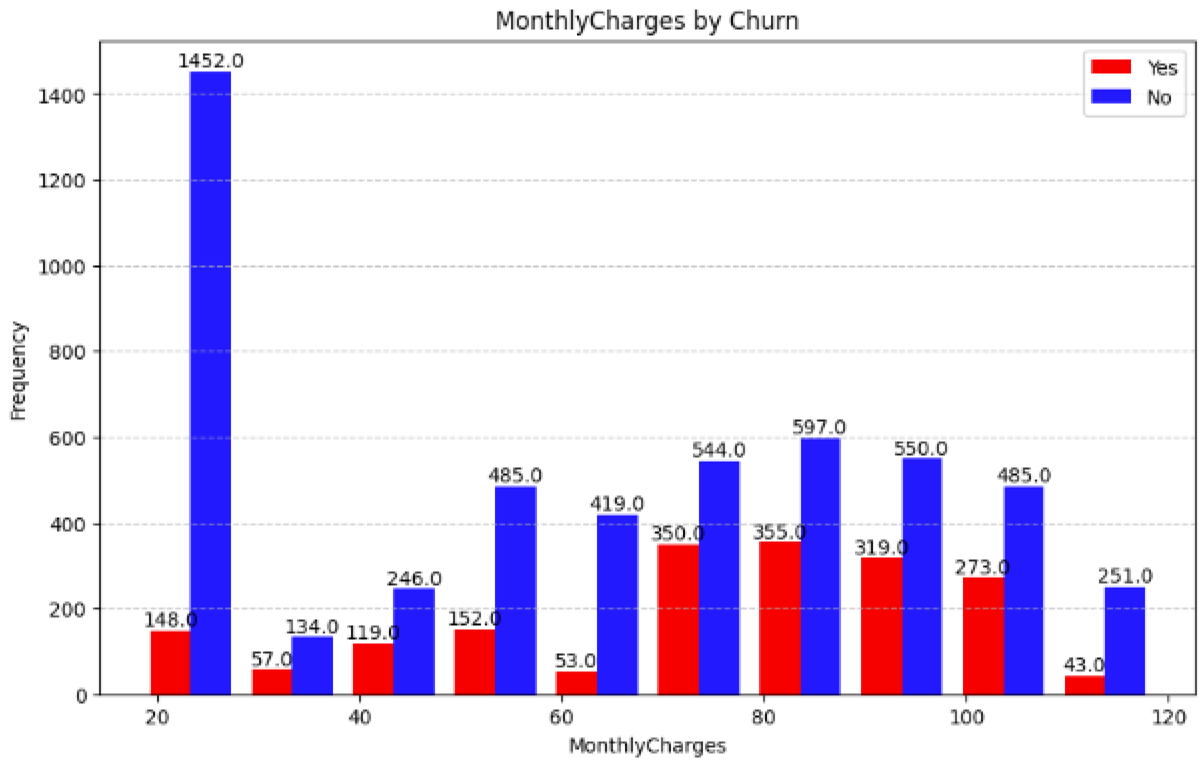
The graph shows that **customers using paperless billing** have a **higher churn rate** compared to those who do not. While most paperless users did not churn, the number of churned users is still relatively large. This suggests that **paperless billing might be associated with increased churn**, potentially due to lower engagement or communication gaps.



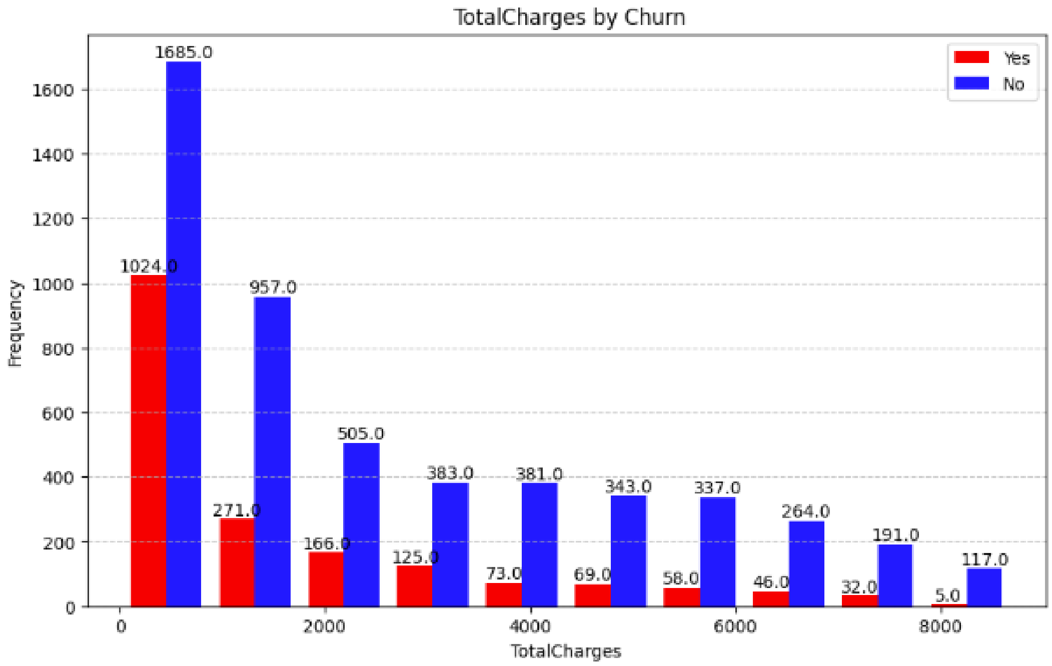
The graph shows that **customers using Electronic check** as a payment method have a **significantly higher churn rate** compared to other methods. In contrast, those who pay via **Bank transfer (automatic)** or **Credit card (automatic)** show **lower churn rates**. This indicates that **automatic payment methods are associated with better customer retention**, while manual methods like electronic checks may contribute to customer dissatisfaction or disengagement.



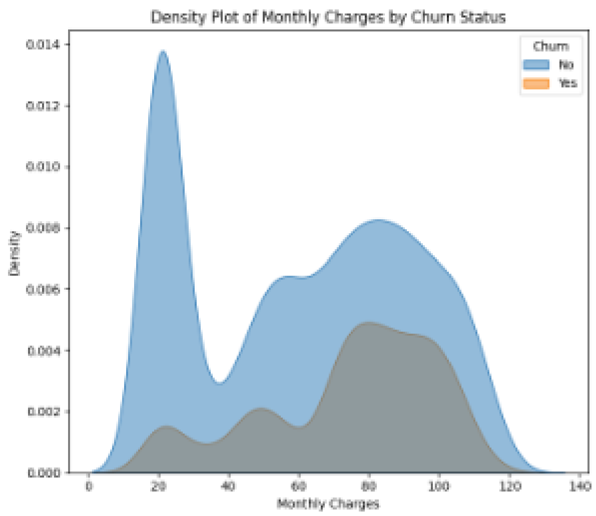
The "**Tenure by Churn**" graph shows that customers with shorter tenure (0–10 months) have the highest churn rate, with over 800 customers leaving the service. As tenure increases, churn significantly decreases, indicating that **long-term customers are more likely to stay**. Particularly, tenure groups beyond 40 months show much higher retention, with churn numbers dropping sharply. The final group (around 70 months) shows a **very high retention rate**, with over 1000 customers not churning. This suggests that **customer loyalty increases with time**, and early engagement strategies may be key to reducing churn.



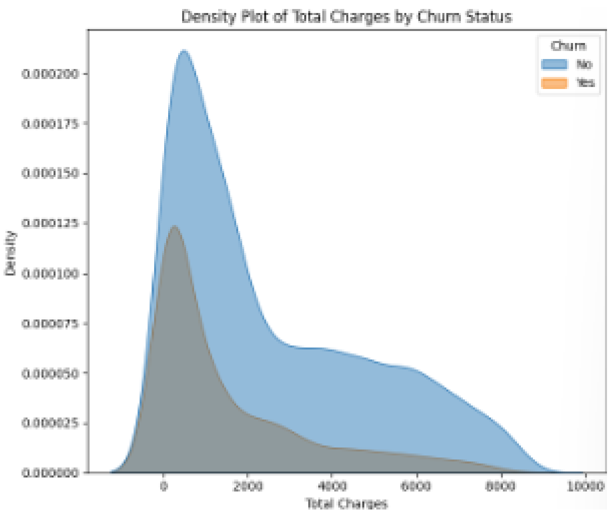
This bar chart shows the distribution of **Monthly Charges** for customers who **churned (red)** and **did not churn (blue)**. The x-axis represents the range of monthly charges, while the y-axis indicates the number of customers in each range. Most customers who did **not churn** had low monthly charges around **$20–$30**, with a very high count (~1450). However, the number of churned customers is more evenly distributed across higher charge ranges, especially between **$60–$100**. This suggests that customers with **higher monthly charges are more likely to churn**, while those with lower charges are more likely to stay.



This chart illustrates how **Total Charges** are distributed among customers who **churned** (red) and those who **did not churn** (blue). Customers with **lower total charges (below $2000)** show a higher churn rate, as seen by the relatively taller red bars in those ranges. In contrast, customers with **higher total charges (above $4000)** are mostly non-churners, indicated by the dominant blue bars. This pattern suggests that **long-term or high-paying customers are more likely to stay**, while **new or low-usage customers are more prone to churn**. It reflects the importance of customer tenure in predicting churn.



This density plot shows the distribution of **Monthly Charges** for customers who **churned (orange)** and **did not churn (blue)**. Customers who did **not churn** are more concentrated at the **lower end (around $20–30)**, while those who **churned** are more spread out across **mid to higher monthly charges**. This indicates that **higher monthly charges are associated with a greater likelihood of churn**.



This density plot compares the **Total Charges** distribution for churned (orange) and non-churned (blue) customers. Churned customers tend to have **lower total charges**, indicating they may have ended service earlier or had shorter tenures. In contrast, customers with **higher total charges** are more likely to have **not churned**, suggesting a link between customer loyalty and long-term spending.

Machine Learning

```
AdaBoost:
Classification Report:
              precision    recall  f1-score   support

         0.0       0.88      0.84      0.86      1034
         1.0       0.60      0.68      0.64       373

    accuracy                           0.80      1407
   macro avg       0.74      0.76      0.75      1407
weighted avg       0.81      0.80      0.80      1407

The AUC-ROC of the model is: 0.845198894425978
```

The classification report for the **AdaBoost model** demonstrates a reasonably strong performance in predicting customer churn. The overall **accuracy of the model is 80%**, which indicates that it correctly classifies a significant majority of the test data. For **non-churned customers** (class 0), the model performs particularly well, achieving a **precision of 0.88** and a **recall of 0.84**, meaning it **accurately identifies most customers who did not churn**. However, the model struggles more with detecting churned customers (class 1), as reflected in the lower **precision of 0.60** and **recall of 0.68**. Despite this, the **AUC-ROC score** of approximately **0.84** suggests that the model has a good overall ability to distinguish between churned and non-churned customers.

```
Gradient Boosting:
Classification Report:
              precision    recall  f1-score   support

         0.0       0.86      0.85      0.86      1034
         1.0       0.61      0.63      0.62       373

    accuracy                           0.79      1407
   macro avg       0.74      0.74      0.74      1407
weighted avg       0.80      0.79      0.79      1407

The AUC-ROC of the model is: 0.8369796879294342
```

The classification report for the **Gradient Boosting model** indicates an overall **accuracy of 79%**, which is slightly better than the AdaBoost model. The model performs well for non-churned customers (class 0), achieving a **precision of 0.86** and a **recall of 0.85**, suggesting it correctly identifies most customers who stay. However, its performance on predicting churned customers (class 1) is more modest, with a **precision of 0.61** and a **recall of 0.63**, indicating a higher number of false negatives. Despite this imbalance, the model shows a strong ability to distinguish between classes, with an **AUC-ROC score of approximately 0.84**, reflecting good overall discrimination capability. This makes Gradient Boosting a competitive model for churn prediction, although there's room to improve recall for the churn class.

```
SVM:
Classification Report:
              precision    recall  f1-score   support

         0.0       0.87      0.82      0.85      1034
         1.0       0.57      0.67      0.62       373

    accuracy                           0.78      1407
   macro avg       0.72      0.75      0.73      1407
weighted avg       0.79      0.78      0.79      1407

The AUC-ROC of the model is: 0.8234724462121642
```

The classification report for the **Support Vector Machine (SVM) model** reveals an overall **accuracy of 78%**, indicating that the model performs reasonably well in predicting customer churn. For non-churned customers (class 0), the model achieves a **precision of 0.87** and a **recall of 0.82**, showing that it is effective at identifying customers who stay. In contrast, the model has lower performance for churned customers (class 1), with a **precision of 0.57** and a **recall of 0.67**, suggesting a tendency to miss a number of true churn cases. The **AUC-ROC score of approximately 0.82** reflects the model's decent ability to differentiate between churn and non-churn classes, though not as strong as Gradient Boosting. Overall, while the SVM model shows balanced accuracy, improvements are needed in detecting actual churners.

```
Logistic Regression:
Classification Report:
              precision    recall  f1-score   support

         0.0       0.91      0.73      0.81      1034
         1.0       0.52      0.80      0.63       373

    accuracy                           0.75      1407
   macro avg       0.71      0.76      0.72      1407
weighted avg       0.81      0.75      0.76      1407

The AUC-ROC of the model is: 0.8359827526304053
```

The **Logistic Regression model** achieved an accuracy of **75%**, indicating reliable overall performance. It performs well in identifying non-churn customers, with a high precision of **0.91** and an F1-score of **0.81**. For churn prediction, the recall is relatively high at **0.76**, but the precision is lower at **0.52**, leading to a moderate F1-score of **0.63**. The model's **AUC-ROC score of 0.84** suggests it is effective in distinguishing between churners and non-churners. However, the lower precision for churn cases indicates a need for improvement, possibly through resampling techniques or more complex models.

```
K-Nearest Neighbors:
Classification Report:
              precision    recall  f1-score   support

         0.0       0.86      0.71      0.78      1032
         1.0       0.46      0.69      0.55       375

    accuracy                           0.70      1407
   macro avg       0.66      0.70      0.67      1407
weighted avg       0.76      0.70      0.72      1407

The AUC-ROC of the model is: 0.7617196382428941
```

The **K-Nearest Neighbors (KNN) model** achieved an overall **accuracy of 70%**, which is lower than that of the logistic regression model. It shows decent performance in predicting non-churn customers, with a precision of **0.86** and an F1-score of **0.78**, but struggles with churn prediction. For churn cases, the precision drops to **0.46**, and the F1-score is **0.55**, indicating many false positives. The **AUC-ROC score of 0.76** suggests moderate discrimination capability between churners and non-churners. Overall, while KNN handles non-churn detection fairly well, it lacks robustness in identifying churned customers accurately.

```
Neural Network:
Classification Report:
              precision    recall  f1-score   support

         0.0       0.83      0.84      0.83      1032
         1.0       0.54      0.53      0.53       375

    accuracy                           0.75      1407
   macro avg       0.69      0.68      0.68      1407
weighted avg       0.75      0.75      0.75      1407

The AUC-ROC of the model is: 0.7875839793281654
```

The **Neural Network model** achieved an **accuracy of 75%**, showing a balanced but modest performance. For non-churn customers, it performs well with a **precision of 0.83** and an **F1-score of 0.83**, but its effectiveness drops for churn cases with a **precision and recall of 0.54 and 0.53**, respectively. The macro average F1-score of **0.68** indicates that the model treats both classes somewhat fairly but struggles with class imbalance. Its **AUC-ROC score is 0.787**, which reflects a reasonable capability in distinguishing between churn and non-churn customers. Overall, the neural network offers a slight improvement over KNN in terms of discrimination, but still needs enhancement in detecting churned customers.

```
Gaussian Naive Bayes:
Classification Report:
              precision    recall  f1-score   support

         0.0       0.91      0.68      0.78      1032
         1.0       0.48      0.81      0.60       375

    accuracy                           0.71      1407
   macro avg       0.69      0.75      0.69      1407
weighted avg       0.79      0.71      0.73      1407

The AUC-ROC of the model is: 0.8169586563307494
```

The **Gaussian Naive Bayes model** achieved an overall **accuracy of 71%**, with a strong recall of **0.81** for churned customers, indicating its effectiveness in identifying customers likely to churn. However, its **precision for churn (class 1) is relatively low at 0.48**, which means it generates more false positives. For non-churn customers, the model performs better, with a **precision of 0.91** but a lower recall of 0.68. The **macro average F1-score of 0.69** and **AUC-ROC of 0.8169** reflect a good balance between classes and reasonable discriminatory ability. Overall, the model is strong in detecting churned customers but could benefit from improved precision for better reliability.

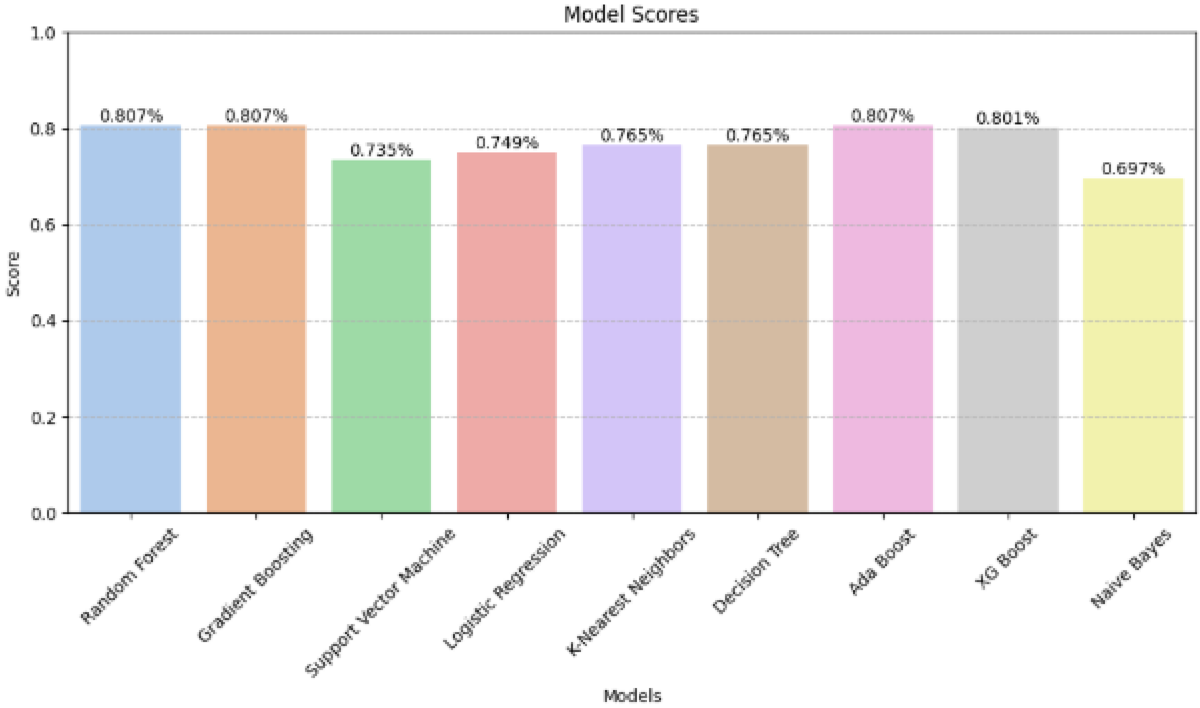
```
Decision Tree classifier:
classification_report:
              precision    recall  f1-score   support

         0.0       0.81      0.90      0.86      1020
         1.0       0.64      0.45      0.53       387

    accuracy                           0.78      1407
   macro avg       0.73      0.68      0.69      1407
weighted avg       0.77      0.78      0.77      1407

The AUC-ROC of the Decision Tree model is: {np.float64(0.8166400668794651)}
```

The **Decision Tree classifier** achieved an overall accuracy of **78%**, showing a relatively strong ability to correctly classify both churn and non-churn cases. The precision for class 0 (non-churn) was **0.81** and recall was **0.90**, indicating the model is very good at identifying customers who did not churn. For class 1 (churn), precision dropped to **0.64** and recall to **0.45**, suggesting it struggles more to correctly identify churners. The macro average F1-score was **0.69**, and the weighted average was **0.77**, showing slightly better performance overall due to class imbalance. The **AUC-ROC score** of **0.8166** reflects decent discrimination between the churn and non-churn classes.

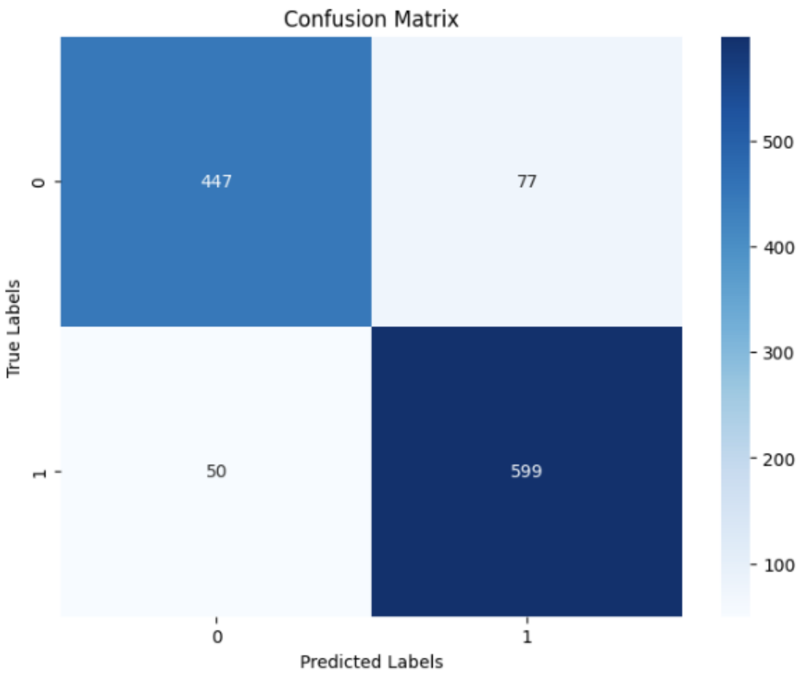
```
Random Forest classifier:
classification_report:
              precision    recall  f1-score   support

         0.0       0.81      0.95      0.87      1034
         1.0       0.73      0.37      0.49       373

    accuracy                           0.80      1407
   macro avg       0.77      0.66      0.68      1407
weighted avg       0.79      0.80      0.77      1407

The AUC-ROC of the Random Forest Classifer model is:  {np.float64(0.8386144543950715)}
```

The **Random Forest classifier** achieved an overall **accuracy of 80%**, performing well in identifying non-churn customers with a **recall of 0.95** and **precision of 0.81** for class 0. However, it struggled with predicting churned customers (class 1), achieving only **0.37 recall** and **0.73 precision**, which led to a lower **F1-score of 0.49**. The macro average F1-score was **0.68**, indicating moderate balance between both classes despite class imbalance. Its **weighted average F1-score of 0.77** reflects good overall model performance. The model's **AUC-ROC score of 0.8386** shows a strong capability in distinguishing between churn and non-churn classes.

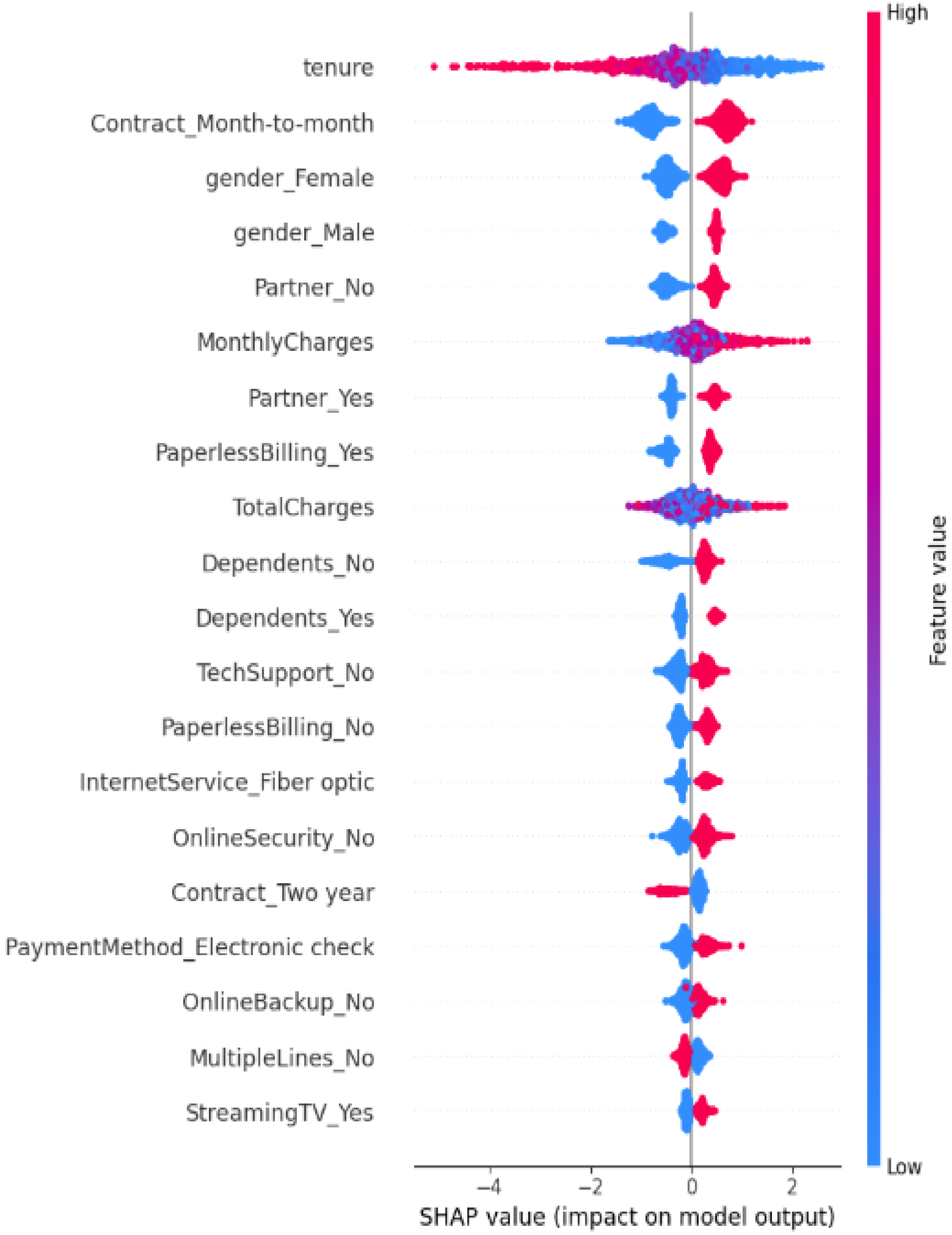
AccuracyChart

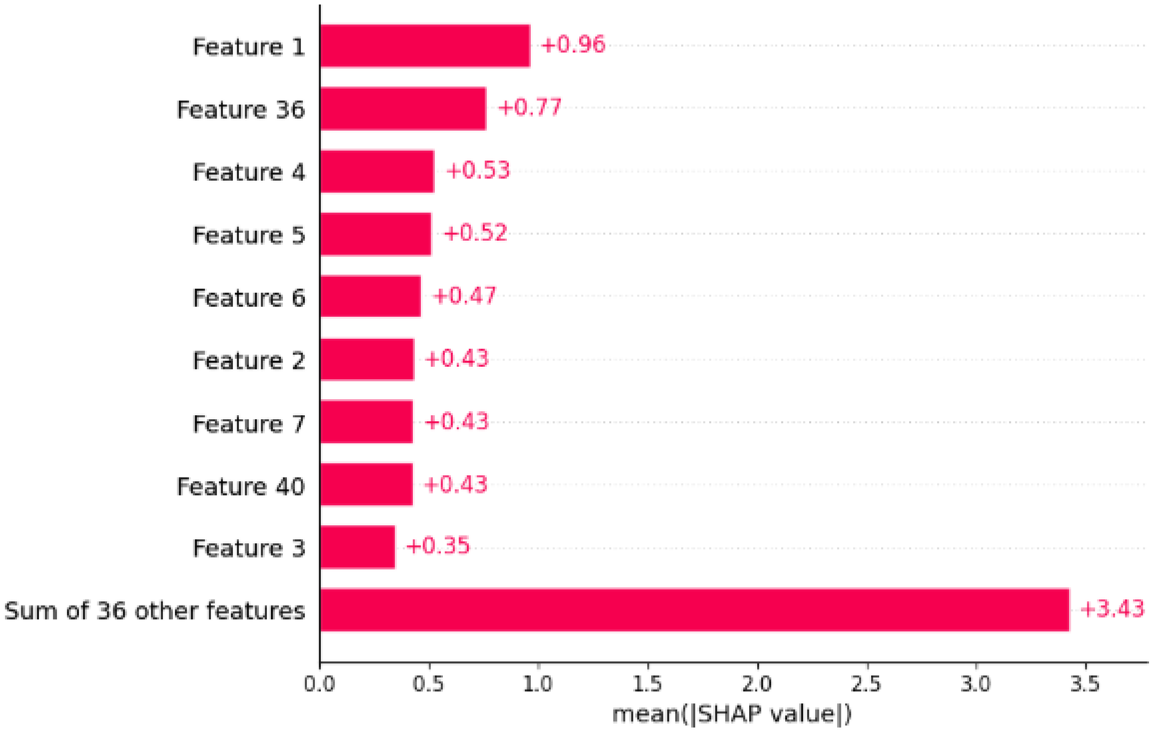


Confusion Matrix



Shap Analysis




Feature 1
+0.96
Feature 36
+0.77
Feature 4
+0.53
Feature 5
+0.52
Feature 6
+0.47
Feature 2
+0.43
Feature 7
+0.43
Feature 40
+0.43
Feature 3
+0.35
Sum of 36 other features
+3.43
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
mean(|SHAP value|)